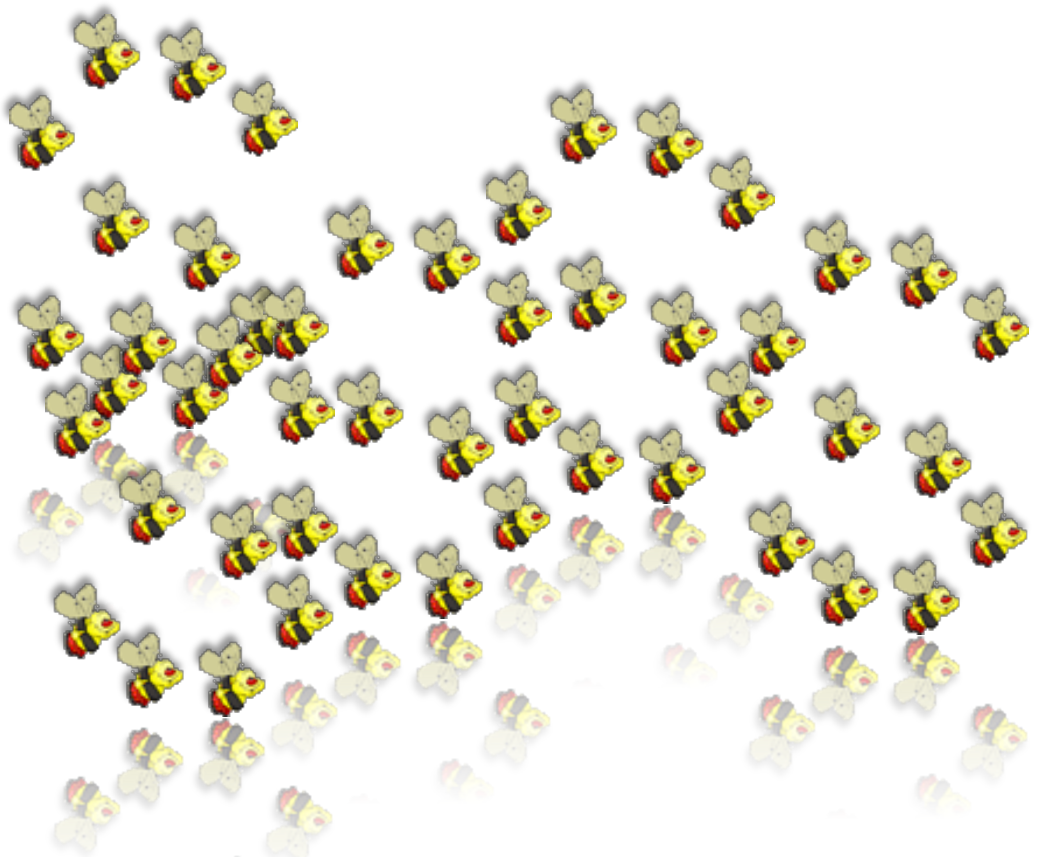


Computational Intelligence II: Evolutionary Optimisation

Chapter 6: Particle Swarm Optimisation



Overview

- ⌘ Particle Swarm Optimisation (PSO) is an optimisation and search algorithm invented by Kennedy & Eberhart in 1995.
- ⌘ Other strategies, such as GAs, GP, or ES (later), mimic the evolutionary process observed in nature through the survival of the fittest and mixing of genetic material.
- ⌘ PSO is also a population-based technique based on some evolutionary analogue, but unlike others it is inspired by social behaviour of animals where all members of the group tend to follow the leader. It does not mix genetic material between individuals.
- ⌘ PSO is based on the fact that social sharing of information among conspecifics offers an evolutionary advantage. It is based on the metaphor of social interaction, such as bird flocking or fish schooling.
- ⌘ It finds the global optima by adjusting the trajectory of each individual toward its own best known location (experience), and the best known location in the entire swarm (sharing).
- ⌘ In PSO, a population is called the **swarm**, while a member is called the **particle**.
- ⌘ PSO can be used to optimise continuous functions more efficiently than other methods.

PSO algorithm

⌘ **Initialisation stage:** To optimise a d-dimensional function $F(x)$, a set of N particles is created in the population (with typically $N=5d$).

- ⊙ Each i^{th} particle is assigned a random position $\mathbf{x}_i = [x_{i1}, \dots, x_{id}]$.
- ⊙ It is also assigned a random velocity vector $\mathbf{v}_i = [v_{i1}, \dots, v_{id}]$.
- ⊙ These two vectors $\mathbf{x}_i$ and $\mathbf{v}_i$ obey some simple bounds.

⌘ **Update stage:** At each generation t , and for each i^{th} particle, its trajectory is changed according to:

$$\begin{aligned}
 v_{ij} &= w \cdot v_{ij} && \leftarrow \text{inertia component} \\
 &+ c_1 \cdot \text{rand} \cdot (pbest_{ij} - x_{ij}) && \text{cognitive component} \rightarrow \\
 &+ c_2 \cdot \text{rand} \cdot (gbest_j - x_{ij}) && \text{social component} \rightarrow \\
 x_{ij} &= x_{ij} + v_{ij}, \quad \forall i = 1, \dots, N, j = 1, \dots, d
 \end{aligned}$$

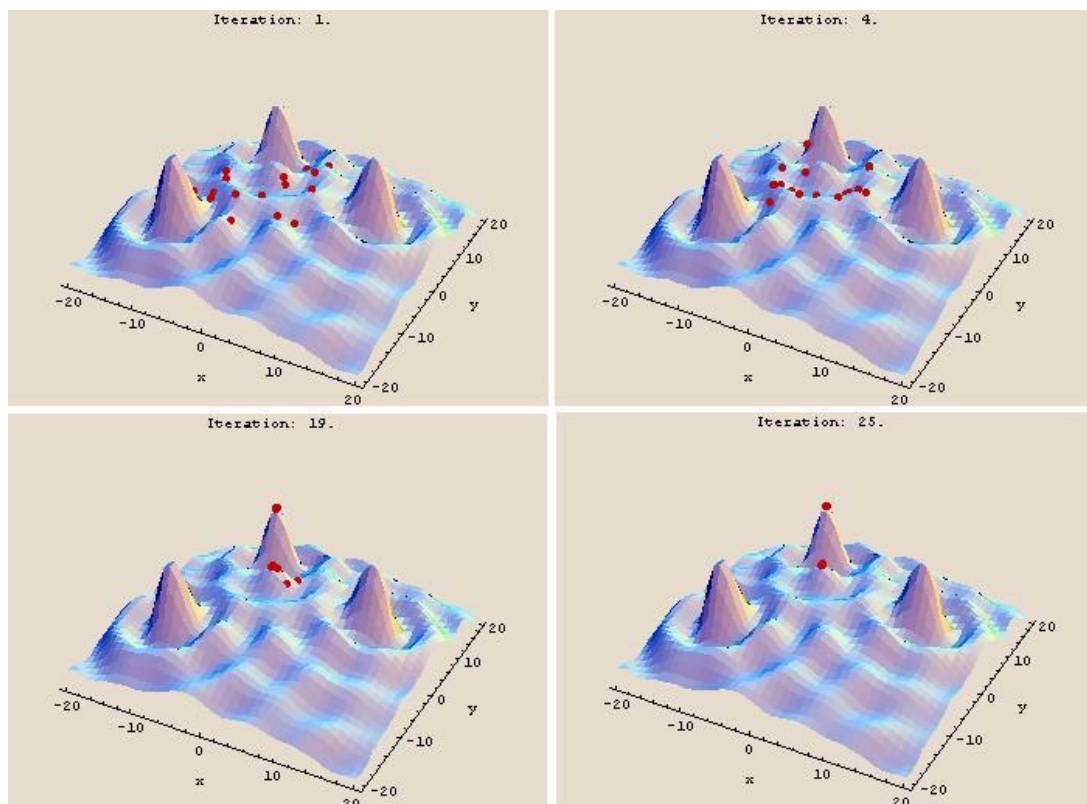
- ⊙ w : the inertia weight $\in [0,1]$, working much like a momentum term in MLPs.
- ⊙ $c_{1\&2}$: some acceleration constants (~ 2.0) that control how far a particle will move in an iteration.
- ⊙ rand : a random uniform number $\in (0,1)$.
- ⊙ x_{ij} : j^{th} position component of i^{th} particle.
- ⊙ v_{ij} : its velocity, maintained within the range $[-v_{\max}, +v_{\max}]$ to reduce the likelihood of particles leaving the useful solution space region.

⌘ Continued...

- ⊙ **pbest_i**: personal best position of the i^{th} particle.
- ⊙ **gbest**: global best position of the entire swarm.
- ⊙ The definition of "best" is based on the optimising (minimising here) objective function $F(x)$. At each generation, the **pbest_i** & **gbest** are kept up-to-date as follows.

$$\text{pbest}_i(t+1) = \begin{cases} \text{pbest}_i(t) & \text{if } F[\mathbf{x}_i(t+1)] \geq F[\text{pbest}_i(t)] \\ \mathbf{x}_i(t+1) & \text{otherwise} \end{cases}$$

$$\text{gbest}(t+1) = \underset{\substack{\text{pbest}_i(t+1) \\ \forall i=1,\dots,n}}{\text{argmin}} F[\text{pbest}_i(t+1)]$$



PSO variations

⌘ There are many PSO variation to enhance the basic scheme presented before, in order to accelerate search and improve efficiency:

⌘ 1} Time Varying Inertia Weight (TVIW) PSO:

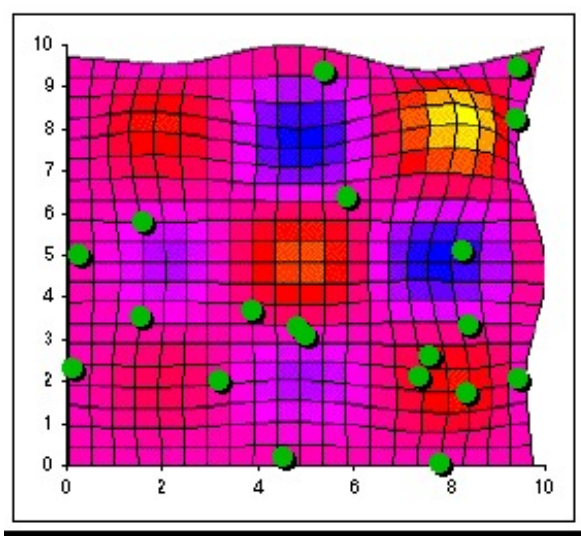
© In the standard velocity update equation, if the inertia w is small (large), then the search facilitate local (global) exploration.

$$v_{ij} = w \cdot v_{ij} + \text{cognitive component} + \text{social component}$$

© This PSO modifies w to vary linearly from w_{init} ($=0.9$) to w_{final} ($=0.4$) during the course of operation to allow better convergence.

$$W = \left(W_{\text{init}} - W_{\text{final}} \right) \frac{t_{\text{final}} - t}{t_{\text{final}}} + W_{\text{final}}$$

© For early (late) generations t , w is large (small), in order to allow more global-search (local fine-tuning).



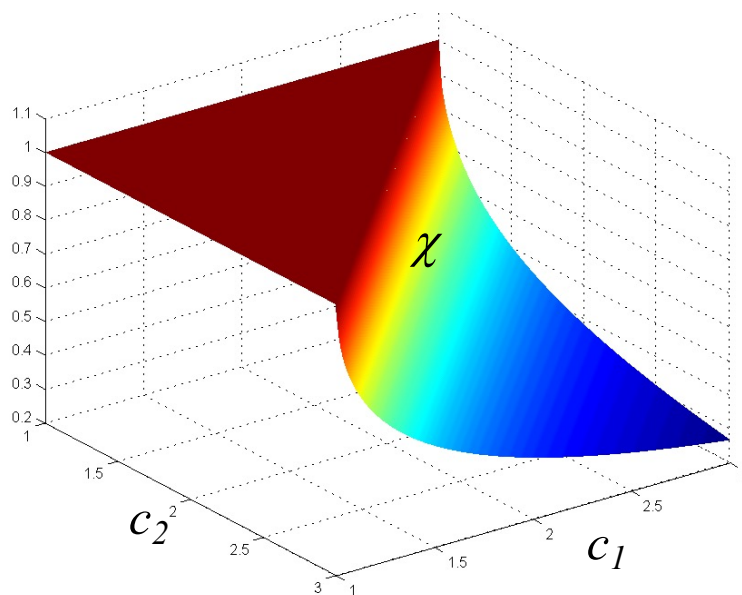
⌘ Continued...

⌘ 2} Constriction Factor (CF) PSO:

- ⊙ This variation helps to ensure better convergence.
- ⊙ It redefines the velocity update equation as

$$v_{ij} = \chi \left\{ \begin{array}{l} v_{ij} + \\ c_1 \cdot \text{rand} \cdot (pbest_{ij} - x_{ij}) + \\ c_2 \cdot \text{rand} \cdot (gbest_j - x_{ij}) \end{array} \right\}, \quad \text{where}$$

$$\chi = \frac{2}{\left| 2 - \varphi - \sqrt{\varphi^2 - 4\varphi} \right|}, \quad \varphi = c_1 + c_2, \varphi > 4$$

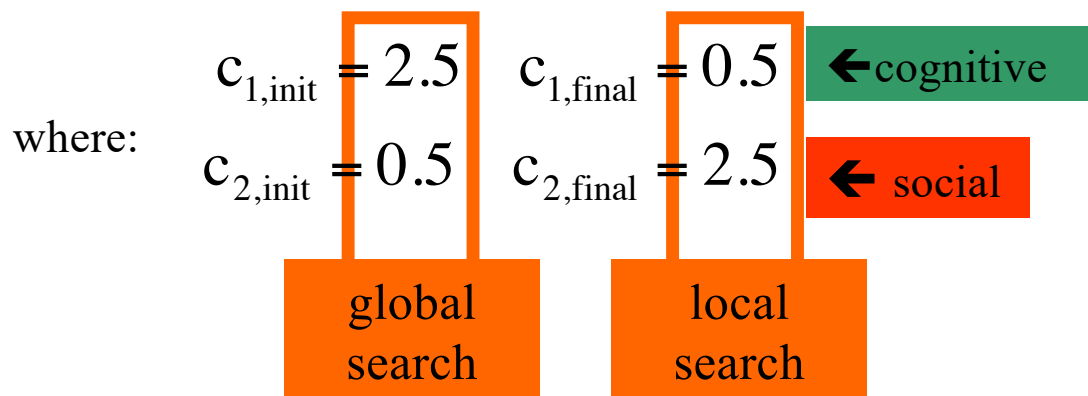


⌘ Continued...

⌘ 3} Time Varying Acceleration Coefficients (TVAC) PSO:

- ⊙ Generally, in population based optimisations, it is beneficial to encourage members to wander through the entire space without clustering around local optima, in the early iterations.
- ⊙ Later on, it is important to support convergence to the global optima.
- ⊙ The TVAC-PSO reduces the cognitive component and increases the social component dynamically, by regulating the acceleration coefficients with time via:

$$c_i = \left(c_{i,\text{final}} - c_{i,\text{init}} \right) \frac{t}{t_{\text{final}}} + c_{i,\text{init}} \quad \text{for } i = 1, 2$$



⌘ Continued...

⌘ 4} Time Varying Acceleration Coefficients with Mutation (TVAC-M) PSO:

- ⊙ Mutation is recognised as an important operator for global search and diversity (e.g., see GA, GP, GEP), so it is introduced here.
- ⊙ When the reached optimum solution is not improving with the progression of generations, a particle is selected randomly and a random perturbation (controlled by a mutation step, which depends on a user-defined quantity m) is added to a random velocity component with a predefined probability mutation rate p_m .
- ⊙ This is implemented as follows:

```
Select a particle k
Select a dimension j
If rate_of_improvement <= threshold, then:
  If rand <  $p_m$ , then
    do:  $v_{kj} := v_{kj} \pm v_{\max} / m$ 
```


⌘ Continued...

⌘ 5} Passive Congregation (PC) PSO:

- ⊙ It improves performance by enhancing (hybridising) the standard PSO as:

$$\begin{aligned}
 V_{ij} = & W \cdot V_{ij} \\
 & + c_1 \cdot \text{rand} \cdot (\text{pbest}_{ij} - x_{ij}) \\
 & + c_2 \cdot \text{rand} \cdot (\text{gbest}_j - x_{ij}) \\
 & + c_3 \cdot \text{rand} \cdot (q_j - x_{ij}) \quad \leftarrow \text{PC component}
 \end{aligned}$$

- ⊙ **q** is a particle selected randomly from the swarm, and c_3 is the passive congregation coefficient.
- ⊙ This PSO, is based on various biologically inspired concepts:
- ⊙ Passive aggregation is a passive grouping of physical processes (e.g., plankton in the sea), while active aggregation is a grouping based on a desirable resources (food, space, etc.).
- ⊙ Congregation is a grouping by social forces. Passive congregation is an attraction of an individual to other members without display of social behaviour.
- ⊙ PC-PSO attempts to minimise missed detection or incorrect interpretation of the solution space by the swarm, by utilising information from other population members effectively.

⌘ Continued...

⌘ 6} LBEST PSO:

- ⊙ It divides the swarm into multiple neighbourhoods, and each neighbourhood maintains its own local best solution.
- ⊙ This solution is less sensitive to local minima effects, at the expense of course of extra computation.

⌘ 7} Social Stereotyping PSO:

- ⊙ This is a variation of the LBEST PSO.
- ⊙ A Clustering algorithm is used to group particles into stereotypical groups.
- ⊙ Each cluster centre $\mathbf{u}_k$ is computed for each k^{th} group, and then used to update the velocity in one of alternative ways:

$$\mathbf{v}_i = w \cdot \mathbf{v}_i + c_1 \cdot \text{rand} \cdot (\mathbf{u}_k - \mathbf{x}_i) + c_2 \cdot \text{rand} \cdot (\mathbf{gbest} - \mathbf{x}_i)$$

$$\mathbf{v}_i = w \cdot \mathbf{v}_i + c_1 \cdot \text{rand} \cdot (\mathbf{pbest}_i - \mathbf{x}_i) + c_2 \cdot \text{rand} \cdot (\mathbf{u}_k - \mathbf{x}_i)$$

$$\mathbf{v}_i = w \cdot \mathbf{v}_i + c_1 \cdot \text{rand} \cdot (\mathbf{u}_k - \mathbf{x}_i) + c_2 \cdot \text{rand} \cdot (\mathbf{u}_k - \mathbf{x}_i)$$

<END of Chapter 6>